Next Steps - Teams Bridge Implementation

Quick Start

This is the implementation roadmap for the **Teams Bridge** service using **Microsoft 365 Agents SDK**.

For complete details, see: ../docs/TEAMS_SDK_INTEGRATION.md

Phase 1: Azure Setup (José Arturo - IT Admin)

Timeline: 1-2 hours
Owner: Azure/Teams Administrator
Status: ☑ COMPLETE

Completed ☑

- ☑ **Register bot in Azure AD**
  - App Name: `ftc-ispilot-corp-prod`
  - Microsoft App ID: `53ead9aa-24aa-44e7-b485-81a988e7492f`
  - Object ID: `f85873ef-78e3-4e6e-97dc-48c4f0e95139`
  - Tenant ID: `c4a8886b-f140-478b-ac47-249555c30afd`
  - Owners: *To be assigned*
- ☑ **Generate Client Secret (App Password)**
  - Secret ID: `b8130e5a-3446-49ef-8267-056e21cabec2`
  - Secret Value: ☑ Stored in .env.example
  - Expiration: 24 months
- ☐ **Configure Teams Bot Channel**
  - Go to Azure Bot Service
  - Create bot resource with Azure credentials above
  - Note: Messaging Endpoint will be: `https://teams-ispilot-bridge-xxxxxx.run.app/api/messages`
  - Enable Teams channel
- ☐ **Verify Azure Permissions**
  - Confirm Azure subscription has Teams integration enabled
  - Verify Microsoft Graph API permissions

Final Deliverables:

- ☑ Microsoft App ID: `53ead9aa-24aa-44e7-b485-81a988e7492f`
- ☑ Microsoft App Secret: ☑ Configured
- ☑ Bot Resource Name: `ftc-ispilot-corp-prod`

- [x] Object ID: `f85873ef-78e3-4e6e-97dc-48c4f0e95139`
- [x] Tenant ID: `c4a8886b-f140-478b-ac47-249555c30afd`
- [x] Secret ID: `b8130e5a-3446-49ef-8267-056e21cabec2`

Phase 2: Build Teams Bridge (Development)

Timeline: 4-6 hours
Owner: Development Team

Prepare

- [ ] Review README.md - Architecture and local setup
- [ ] Review ../docs/TEAMS_SDK_INTEGRATION.md - Complete guide
- [ ] Have Azure credentials from Phase 1 ready

Implementation

1. **Setup development environment**

```
cd teams_bot_bridge
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

2. **Update main.py**
   - [ ] Import MS 365 Agents SDK modules
   - [ ] Implement JWT validation for Azure Bot Service
   - [ ] Add identity token generation for sa-tot-osa
   - [ ] Implement message routing to ispilot-api
   - [ ] Add session context management
   - See: ../docs/TEAMS_SDK_INTEGRATION.md → "Phase 2: Build Teams Bridge"

3. **Create Dockerfile**
   - [ ] Copy or update from template
   - [ ] Ensure all dependencies in requirements.txt

4. **Create Teams app manifest** (manifest.json)
   - [ ] Configure app details
   - [ ] Set scopes and capabilities
   - See: ../docs/TEAMS_SDK_INTEGRATION.md → "Teams App Manifest"

5. **Test locally**
   - [ ] Run: `uvicorn main:app --reload`
   - [ ] Health check: `curl http://localhost:8080/health`

    - ☐ Review startup logs

Deliverables:

- ☐ main.py implementing MS 365 Agents SDK
- ☐ Updated requirements.txt with all dependencies
- ☐ Dockerfile (production-ready)
- ☐ manifest.json for Teams app
- ☐ .env.example with all variables
- ☐ Local testing confirmed working

Phase 3: Deploy to Cloud Run

Timeline: 1-2 hours
Owner: Development Team (GCP Admin for IAM)

Pre-Deployment Checks

- ☐ Verify all files are in place (main.py, Dockerfile, requirements.txt)
- ☐ Confirm ispilot-api is running and accessible
- ☐ Test identity token generation locally:

```
gcloud auth print-identity-token \
  --impersonate-service-account=sa-tot-osa@corp-stro-salesinventory-
prod.iam.gserviceaccount.com
```

Deploy

1. **Deploy to Cloud Run**

```
cd teams_bot_bridge

PROJECT_ID="corp-stro-salesinventory-prod"
REGION="us-central1"
SERVICE_NAME="teams-ispilot-bridge"
SERVICE_ACCOUNT="sa-tot-osa@${PROJECT_ID}.iam.gserviceaccount.com"

gcloud run deploy "${SERVICE_NAME}" \
  --source . \
  --region "${REGION}" \
  --project "${PROJECT_ID}" \
  --service-account "${SERVICE_ACCOUNT}" \
  --allow-unauthenticated \
  --memory 512Mi \
  --cpu 1 \
  --timeout 60 \
  --set-env-vars \
```

```
    MICROSOFT_APP_ID=<APP_ID>,\
    MICROSOFT_APP_PASSWORD=<APP_PASSWORD>,\
    ISPILOT_API_ENDPOINT=https://ispilot-api-46y2f3tyja-
uc.a.run.app/chat,\
    GOOGLE_CLOUD_PROJECT=${PROJECT_ID}
```

2. **Get Service URL**

```
gcloud run services describe teams-ispilot-bridge \
  --region=us-central1 \
  --project=corp-stro-salesinventory-prod \
  --format='value(status.url)'
```

3. **Verify Deployment**

   - ☐ Service is running: `gcloud run services describe teams-ispilot-bridge --region=us-central1`
   - ☐ Health check: `curl https://<SERVICE-URL>/health`
   - ☐ Check logs: `gcloud run logs read teams-ispilot-bridge --region=us-central1 --limit 20`

Deliverables:

- ☐ Cloud Run service deployed and running
- ☐ Service URL obtained
- ☐ Environment variables configured
- ☐ Health endpoint responding

Phase 4: Configure Azure & Teams

Timeline: 1-2 hours
Owner: Azure/Teams Admin + Development

Setup Azure Bot Service

- ☐ **Set Messaging Endpoint**
  - Go to Azure Bot Service → Configuration
  - Set Messaging Endpoint to: `https://<SERVICE-URL>/api/messages`
  - Save and verify
- ☐ **Test Bot in Emulator** (optional)
  - Download Bot Framework Emulator
  - Connect with Azure bot credentials
  - Send test message

Publish Teams App

- [ ] **Update manifest.json** with service URL and bot ID
- [ ] **Test in Teams Web Client**
  - Or publish to Teams App Catalog
- [ ] **Add app to team/group chat**
- [ ] **Send test message**
  - Expected: Bot responds with Reasoning Engine output

Deliverables:

- [ ] Azure messaging endpoint configured
- [ ] Teams app manifest updated
- [ ] App added to Teams test team/chat
- [ ] End-to-end message flow verified

Phase 5: Validation & Go-Live

Timeline: 2-4 hours
Owner: Development Team + QA

Validation Checklist

Functional Testing

- [ ] Single-turn conversation works (send message, get response)
- [ ] Multi-turn conversation works (session continuity)
- [ ] Different message types handled (text, etc.)
- [ ] Error cases handled gracefully (timeouts, API errors)

Integration Testing

- [ ] Teams activity validation (correct JWT)
- [ ] Identity token generation works
- [ ] ispilot-api integration works end-to-end
- [ ] Session context persisted correctly

Performance Testing

- [ ] Response time < 30 seconds (typical)
- [ ] 100+ concurrent requests handled
- [ ] No memory leaks or resource exhaustion
- [ ] Cloud Run logs show healthy operation

Deployment Ready Checks

- [ ] All GitHub commits pushed
- [ ] Documentation updated
- [ ] Team trained on bot usage

- ☐ Support runbook created
- ☐ Monitoring/logging configured
- ☐ Rollback plan documented

Deliverables:

- ☐ All validation tests passing
- ☐ Performance benchmarks documented
- ☐ User documentation ready
- ☐ Support procedures established
- ☐ Ready for production rollout

Quick Reference

Key Files

File	Purpose	Status
main.py	Teams Bridge implementation	☑ Complete
requirements.txt	Python dependencies	☑ Complete
Dockerfile	Container definition	☑ Complete
manifest.json	Teams app manifest	☑ Complete
.env.example	Environment template	☑ Complete
deploy.sh	Cloud Run deployment	☑ Complete
README.md	Architecture & setup	☑ Complete
NEXT_STEPS.md	This file	☑ Complete

Critical Environment Variables

```
# Required for Phase 1
MICROSOFT_APP_ID=<from-azure-registration>
MICROSOFT_APP_PASSWORD=<from-azure-registration>

# Auto-configured
GOOGLE_CLOUD_PROJECT=corp-stro-salesinventory-prod
ISPILOT_API_ENDPOINT=https://ispilot-api-46y2f3tyja-uc.a.run.app/chat
```

Key API Endpoints

Endpoint	Method	Purpose
/health	GET	Health check

Endpoint	Method	Purpose
/api/messages	POST	Handle Teams activities
/api/activities	POST	Alternative handler (same)

Troubleshooting

Issue: "Identity token generation failed"
Solution: Verify sa-tot-osa service account exists and has Workload Identity bindings

Issue: "JWT validation failed"
Solution: Ensure MICROSOFT_APP_ID and MICROSOFT_APP_PASSWORD are correct from Azure

Issue: "ispilot-api returned 401"
Solution: Check identity token generation; verify sa-tot-osa has required permissions on ispilot-api

Issue: "No message text found in activity"
Solution: Verify Teams activity payload format; check TeamsActivityParser.extract_message() logic

Support Contacts

- **Azure/Teams Setup**: José Arturo (Phase 1)
- **Development**: Development Team (Phase 2)
- **Deployment**: GCP Admin (Phase 3)
- **Integration**: DevOps Team (Phase 4)
- **Testing/QA**: QA Team (Phase 5)

Success Criteria

☑ **Teams Bridge is ready when:**

- All phases completed
- All validation tests passing
- Response times meet SLA (< 30 seconds)
- Zero critical bugs
- Documentation complete
- Team trained and confident

Estimated Total Duration: 12-16 hours across all phases

Last Updated: 2026-09-01
Version: 1.0.0
Status: Ready for Phase 1 execution

- ☐ Cloud Run logs show no errors
- ☐ Monitor resource usage (CPU, memory)

Security Validation

- [ ] JWT validation rejects invalid tokens
- [ ] Identity tokens generated properly
- [ ] ispilot-api accepts only valid tokens
- [ ] No secrets in logs

Go-Live Readiness

- [ ] All tests passing
- [ ] Documentation updated
- [ ] Team trained on new interface
- [ ] Rollback plan documented
- [ ] Support team notified

Deliverables:

- [ ] Test report
- [ ] Performance baseline
- [ ] Runbook for support team
- [ ] Go/No-go decision

Timeline Overview

Phase	Duration	Status
Phase 1: Azure Setup	1-2 hrs	⏳ Waiting for José Arturo
Phase 2: Build Bridge	4-6 hrs	⏳ Blocked on Phase 1
Phase 3: Cloud Deploy	1-2 hrs	⏳ After Phase 2
Phase 4: Teams Config	1-2 hrs	⏳ After Phase 3
Phase 5: Validation	2-4 hrs	⏳ After Phase 4
Total	**~12-16 hrs**	-

Resources

Documentation

- README.md - Quick reference
- ../docs/TEAMS_SDK_INTEGRATION.md - Complete implementation guide
- ../docs/ARCHITECTURE_RESET_TEAMS_SDK.md - Architecture decisions
- ../ispilot-api/README.md - ispilot-api reference

Code Templates

- main.py template in TEAMS_SDK_INTEGRATION.md
- Dockerfile in this directory

- manifest.json in this directory (needs update)

External Links

- Microsoft 365 Agents SDK
- Azure Bot Service
- Teams App Manifest

Questions?

Refer to:

1. ../docs/TEAMS_SDK_INTEGRATION.md - Complete guide with detailed instructions
2. README.md - Troubleshooting section
3. ../WORK_CHECKPOINT.md - Current implementation status